

MODULE – III COMPLETE ANSWERS (DBMS)

Based on the uploaded question bank.

SHORT ANSWER QUESTIONS (5 MARKS)

1. Explain Data Definition Language (DDL) with Suitable Commands

DDL (Data Definition Language) is used to define and modify the database structure.

Common DDL Commands

CREATE

Creates a database object.

```
CREATE TABLE Student(  
SID INT,  
Name VARCHAR(30),  
Dept VARCHAR(20)  
);
```

ALTER

Modifies table structure.

```
ALTER TABLE Student  
ADD Age INT;
```

DROP

Deletes a table.

DROP TABLE Student;

TRUNCATE

Removes all records.

TRUNCATE TABLE Student;

Characteristics

- Defines schema.
 - Automatically commits changes.
 - Changes database structure.
-

2. SQL Query to Display Minimum, Maximum, Average and Total Salaries

Assume table:

Employee(EmplID, Name, Salary)

Minimum Salary

```
SELECT MIN(Salary) AS Minimum_Salary  
FROM Employee;
```

Maximum Salary

```
SELECT MAX(Salary) AS Maximum_Salary  
FROM Employee;
```

Average Salary

```
SELECT AVG(Salary) AS Average_Salary  
FROM Employee;
```

Total Salary

```
SELECT SUM(Salary) AS Total_Salary  
FROM Employee;
```

Combined Query

```
SELECT  
MIN(Salary) AS Minimum,  
MAX(Salary) AS Maximum,  
AVG(Salary) AS Average,  
SUM(Salary) AS Total  
FROM Employee;
```

3. Explain Data Manipulation Language (DML)

DML is used to manipulate data stored in database tables.

Commands

INSERT

```
INSERT INTO Student  
VALUES(101,'Anik','CSE');
```

UPDATE

```
UPDATE Student  
SET Dept='IT'  
WHERE SID=101;
```

DELETE

```
DELETE FROM Student  
WHERE SID=101;
```

SELECT

```
SELECT * FROM Student;
```

Advantages

- Data insertion
- Data modification

- Data deletion
 - Data retrieval
-

4. Explain Projection Operation in Relational Algebra

Projection selects specific columns from a relation.

Symbol

π

Syntax

π Attribute_List(Relation)

Example

STUDENT

SID	Name	Dept
101	Anik	CSE
102	Rahul	EE

Query:

π Name(STUDENT)

Result

Name

Anik

Rahul

Purpose

Removes unwanted columns.

5. Difference Between Cartesian Product and Natural Join

Cartesian Product

Combines every tuple of one relation with every tuple of another relation.

Symbol

x

Example

R

A

1

2

S

B

X

Y

Result:

A B

1 X

1 Y

2 X

2 Y

Total = 4 tuples

Natural Join

Combines tuples based on common attributes.

Symbol

⋈

Example

Student

SID	DeptID
-----	--------

1	10
---	----

Department

DeptID	DName
--------	-------

10	CSE
----	-----

Result

SID	DeptID	DName
-----	--------	-------

1	10	CSE
---	----	-----

Difference

Cartesian Product

Natural Join

No condition

Uses common attribute

Large result

Meaningful result

$\times$ operator

$\bowtie$ operator

6. Example of Rename Operation in Relational Algebra

Rename operation changes relation or attribute names.

Symbol

ρ

Syntax

$\rho(\text{NewRelation}, \text{OldRelation})$

Example

$\rho(\text{Employee}, \text{Staff})$

Meaning:

Rename Staff relation as Employee.

Attribute Rename

$\rho(\text{SID}, \text{Name}, \text{Dept})$
(STUDENT)

Used to avoid ambiguity.

7. Explain

(i) DDL

Used to define database structure.

Commands:

```
CREATE  
ALTER  
DROP  
TRUNCATE
```

(ii) DCL

Data Control Language controls access rights.

Commands:

```
GRANT  
REVOKE
```

Example:

```
GRANT SELECT  
ON Student  
TO User1;
```

(iii) Views

A view is a virtual table derived from one or more tables.

Example:

```
CREATE VIEW CSEStudents AS  
SELECT *  
FROM Student  
WHERE Dept='CSE';
```

8. What is a View? How Can It Be Created?

A view is a virtual table whose contents are obtained from another table.

Advantages

- Security
 - Simplicity
 - Data abstraction
-

Syntax

```
CREATE VIEW view_name AS  
SELECT columns  
FROM table_name;
```

Example

```
CREATE VIEW EmployeeView AS  
SELECT EmpID, Name  
FROM Employee;
```

Usage

```
SELECT * FROM EmployeeView;
```

9. What is Join Operator? Explain Conditional Join and Natural Join

A Join combines rows from multiple relations.

Conditional Join (Theta Join)

Uses comparison operators.

Syntax

R ⋈ condition S

Example

Employee ⋈ Employee.DeptID=Department.DeptID Department

Natural Join

Automatically joins using common attributes.

Syntax

Employee ⋈ Department

Example

Employee

EmpID	DeptID
1	10

Department

DeptID	Name
10	CSE

Result

EmpID	DeptID	Name
1	10	CSE

10. How Can We Compare Using NULL Values? Explain Logical Connectives

NULL means unknown or missing value.

Checking NULL

```
SELECT *  
FROM Employee  
WHERE Salary IS NULL;
```

NOT NULL

```
SELECT *  
FROM Employee  
WHERE Salary IS NOT NULL;
```

Logical Connectives

AND

```
SELECT *  
FROM Employee  
WHERE Dept='CSE'  
AND Salary>50000;
```

Both conditions must be true.

OR

```
SELECT *  
FROM Employee  
WHERE Dept='CSE'  
OR Dept='EE';
```

At least one condition true.

NOT

```
SELECT *  
FROM Employee  
WHERE NOT Dept='CSE';
```

Negates condition.

LONG ANSWER QUESTIONS (10 MARKS)

1. Discuss Different Types of Outer Join with Example

Outer joins preserve unmatched rows.

Left Outer Join

Returns all rows from left table.

```
SELECT *  
FROM Employee E  
LEFT JOIN Department D  
ON E.DeptID=D.DeptID;
```

Right Outer Join

Returns all rows from right table.

```
SELECT *  
FROM Employee E  
RIGHT JOIN Department D  
ON E.DeptID=D.DeptID;
```

Full Outer Join

Returns all matching and non-matching rows.

```
SELECT *  
FROM Employee E  
FULL OUTER JOIN Department D
```

ON E.DeptID=D.DeptID;

Comparison

Join	Preserves
Left	Left Table
Right	Right Table
Full	Both Tables

2. Discuss Different Types of Integrity Constraints

Integrity constraints ensure correctness of data.

Domain Constraint

Valid values only.

Example:

Age > 0

Key Constraint

Primary key must be unique.

Example:

SID UNIQUE

Entity Integrity

Primary key cannot be NULL.

Referential Integrity

Foreign key references valid primary key.

Example

Department(DeptID)

Employee(
EmpID,
DeptID
)

DeptID must exist in Department.

3(i). Relational Algebra for Regno and Section where Branch='EE'

Relation:

Students(Regno, Name, Branch, Section)

Selection

σ Branch='EE' (Students)

Projection

π Regno, Section
(σ Branch='EE'(Students))

3(ii). Relational Algebra for Books where Topic='Database' and Author='Ram'

Relation:

Book(BookID,Title,Topic,Author)

Answer:

σ Topic='Database'
 $\wedge$ Author='Ram'
(Book)

4. What are Aggregate Functions? Explain Five Built-in Aggregate Functions

Aggregate functions operate on groups of rows.

COUNT()

Counts rows.

```
SELECT COUNT(*)  
FROM Employee;
```

SUM()

Calculates total.

```
SELECT SUM(Salary)  
FROM Employee;
```

AVG()

Calculates average.

```
SELECT AVG(Salary)
FROM Employee;
```

MIN()

Returns smallest value.

```
SELECT MIN(Salary)
FROM Employee;
```

MAX()

Returns largest value.

```
SELECT MAX(Salary)
FROM Employee;
```

5. Structure of Basic SQL Query and Cursor

Basic SQL Query Structure

```
SELECT column_names
FROM table_name
WHERE condition;
```

Example:

```
SELECT Name
FROM Student
WHERE Dept='CSE';
```

Cursor

A cursor is a database object used to process records one at a time.

User Defined Cursor

```
DECLARE ManagerCursor CURSOR  
FOR  
SELECT *  
FROM Employee  
WHERE Designation='Manager';
```

Open Cursor

```
OPEN ManagerCursor;
```

Fetch Data

```
FETCH NEXT  
FROM ManagerCursor;
```

Close Cursor

```
CLOSE ManagerCursor;
```

6. Relational Algebra Expressions

Relations

STUDENT(SID,Name,Dept,Age)

COURSE(CID,CName,Dept)

ENROLL(SID,CID,Marks)

Names of all students

π Name(STUDENT)

Students whose age > 20

π Name
(σ Age>20(STUDENT))

Students enrolled in DBMS

π Name
(
STUDENT $\bowtie$ ENROLL $\bowtie$
 σ CName='DBMS'(COURSE)
)

Students not enrolled in any course

π SID(STUDENT)
-
 π SID(ENROLL)

Students enrolled in all courses

Division operator:

π SID,CID(ENROLL)
 $\div$
 π CID(COURSE)

7. Difference Between Natural Join and Cross Product

Natural Join

Uses common attribute

Cross Product

No condition

Meaningful result

All combinations

Smaller relation

Larger relation

Symbol ⋈

Symbol ×

Example

Employee

EID	DeptID
-----	--------

1	10
---	----

Department

DeptID	Name
--------	------

10	CSE
----	-----

Natural Join

Employee ⋈ Department

Result:

EID	DeptID	Name
-----	--------	------

1	10	CSE
---	----	-----

Cross Product

Employee × Department

All combinations generated.

Types of Joins

(a) Natural Join

Uses common attribute automatically.

$R \bowtie S$

(b) Equi Join

Uses equality condition.

$R \bowtie R.A=S.A S$

(c) Self Join

Table joined with itself.

```
SELECT *  
FROM Employee E1,  
Employee E2;
```

(d) Inner Join

Returns matching tuples only.

```
SELECT *  
FROM Employee  
INNER JOIN Department  
ON Employee.DeptID=  
Department.DeptID;
```

(e) Outer Join

Returns matching and unmatched tuples.

Types:

- Left Outer Join
- Right Outer Join

- Full Outer Join
-

EXAM REVISION (MOST IMPORTANT)

1. DDL Commands
2. DML Commands
3. Aggregate Functions
4. Views
5. Natural Join
6. Outer Join
7. Relational Algebra Operators
8. Projection and Selection
9. Cursor
10. SQL Queries

These topics are asked repeatedly in university examinations.